Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 4 · SLO 1 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 4 : SLO â€“ 1 | Array Implementation of Lists â€” Programming Practice

I. Max and Min

• int max = a[0], min = a[0];

• for (i = 1; i < n; i++) { if (a[i] > max) max = a[i]; if (a[i] < min) min = a[i]; }

II. Frequency of Each Element

• for (i = 0; i < n; i++) { if (counted[i]) continue; int c = 1;

• for (j = i + 1; j < n; j++) if (a[j] == a[i]) { c++; counted[j] = 1; }

• printf("%d occurs %d times\n", a[i], c); counted[i] = 1; }

III. Separate Negatives, Positives, Zeroes

• for (i = 0; i < n; i++) {

• if (a[i] < 0) neg[nn++] = a[i];

• else if (a[i] > 0) pos[np++] = a[i];

• else zero[nz++] = a[i]; }

IV. Reverse the Array

• for (i = 0, j = n - 1; i < j; i++, j--) { int t = a[i]; a[i] = a[j]; a[j] = t; }

V. Print Odd Numbers

• for (i = 0; i < n; i++) if (a[i] % 2 != 0) printf("%d ", a[i]);